Instructions

There are **50 total points**. This exam is closed-book and closed-notes except for **one 8.5 × 11-inch sheet of paper with hand-written notes (both sides permitted)**. No electronic devices, additional pages, or other materials are permitted. When asked to provide your answer within a figure or table, be careful to not exceed box boundaries. Bubbles must be filled out completely: ● is correct, ✓ ⊙ ☒ are incorrect . Write your full name in the box to acknowledge the instructions.

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Motivation and Applications

1. [3 points]: Since most streaming video became encrypted, which of the following can a network operator *no longer read directly* from the traffic, and so must infer? Select all that apply.

- [x] Video resolution
- [x] Rebuffering (stall) events
- [x] Startup delay before playback
- [ ] Packet sizes and timing
- [ ] IP addresses of the video servers
- [ ] Number of parallel connections

2. [2 points]: In Hands-On 1 you looked at a packet capture before doing anything else with it. Which observations should make you suspect a problem with the data? Select all that apply.

- [x] A packet with a negative length
- [ ] A packet far above the maximum packet size
- [x] Timestamps that go backwards
- [ ] Large packets flow mostly server to client
- [ ] A handshake before the large packets
- [ ] Round-trip times that vary over the trace

3. [3 points]: Attackers are preparing a phishing campaign that imitates the University's login page. You run the campus network and can obtain domain registration data and the logs of the campus DNS resolver. Name **one** data source and **two** features you would use to detect the campaign, ideally before the first email arrives.

(Answer inside the box)

Look at data domain registration related to UChicago and measure network traffic from them. I would train on how emails were sent from the same source, and how frequently they come, # recipients

4. [3 points]: In Hands-On 2, the Log4j scan and the normal web trace were captured at different times of day, so a model with time of day as a feature separates them perfectly. Is time of day a good feature for detecting scans? ○ Yes ● No

Why or why not?

(Answer inside the box)

It is a spurious feature. You could get an attack during the day and completely miss it.

Data Acquisition

5. [3 points]: Which of the following are *passive* measurements? Select all that apply.

- [x] An ISP collects flow records from its routers
- [x] Campus logs every query its DNS resolver answers
- [ ] Inferring resolution from a capture of your Netflix session
- [ ] An app downloads a test file every hour
- [ ] A laptop pings a server once a minute
- [ ] Measuring latency while a test upload runs

Initials: AB

6. [3 points]: A student reports that Zoom was choppy half an hour ago. You run a speed test from their laptop now and it reports high throughput and low latency. Does this show the network was not the cause? ◯ Yes ◉ No

Why or why not?

(Answer inside the box)

Likely that it is a temporary ~~boost~~ Stake. Zoom servers or internet could have been more busy 30 mins ago.

7. [3 points]: (a) Name the five header fields that define a flow. In Hands-On 5 it was safe to group packets by only four of them; which one could be left out, and why? (b) Two packets match on all five fields but arrive 45 seconds apart with no packets in between. Would a router using conventional settings put them in the same flow record? Why or why not?

(Answer inside the box)

a) Src/Dst IP, Src/Dst port and protocol. You had to leave out dst port because it could vary a ton and create inaccurate flows
b) It would not because generally, groups flow or a 30s window.

Features and Representation

8. [4 points]: In Hands-On 6 you used `netml` to turn a capture into per-flow STATS features. nPrint instead turns each packet into a fixed-width bitmap of its header bits and lets a model learn which bits matter. Give **one advantage** and **one disadvantage** of *each* approach.

(Answer inside the box)

With STATS, you are more likely to get features that matter to the objective due to domain knowledge. However, you lose byte level granularity. With nPrint, you can train much more detailed through by far BUT you could learn spurious features.

9. [2 points]: A flow's protocol is TCP, UDP, or ICMP, and you encode it as 0, 1, 2. Which statements are true? Select all that apply.

- ◉ This is usually fine for a random forest
- ◉ A linear model treats ICMP as "twice" UDP
- ◉ One-hot encoding gives three binary columns
- ◯ Min-max scaling fixes it for a linear model
- ◯ Tree models require one-hot inputs
- ◯ The order 0, 1, 2 carries real meaning

10. [3 points]: Some rows in your flow dataset have missing values. Give **one** example from network data where filling in the column average is reasonable, and **one** where it is a bad idea. Explain why for the second.

(Answer inside the box)

This is fine when your dataset has only a couple missing and it's not a metric you're trying to predict. If it's something you are trying to predict, it's a really bad idea.

Training and Evaluation

11. [3 points]: You have six months of hourly traffic volumes and want to forecast next month's load to decide whether to add a server. You shuffle all rows randomly, split 80/20, and get low test error. Is that a trustworthy estimate of how well the model will forecast? ○ Yes ● No

Why or why not?

(Answer inside the box)

You could have gotten lucky with the split and training data represent test well but doesn't imply it will do well in production

12. [5 points]: A detector labels 1,000 flows; 20 of them are attacks (attack is the positive class). Its confusion matrix:

	Predicted attack	Predicted benign
True attack	15	5
True benign	15	965

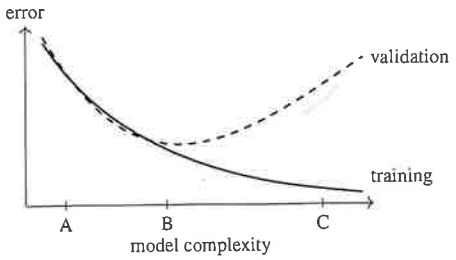
(a) Compute accuracy, precision, and recall. (b) What is the accuracy of a classifier that always says "benign"? What does the comparison tell you, and which curve would you use to compare detectors on this data?

(Answer inside the box)

a) accuracy = 980/1000 = 98%. Precision = 15/30 = 50% Recall = 75%

b) 98%. When attacks are rare, you want to use precision vs Recall graphs.

13. [3 points]: The figure shows training and validation error as model complexity increases.



(a) Which point (A, B, C) has high bias? (b) Which is overfitting? (c) Name one way to reduce overfitting other than choosing a simpler model.

(Answer inside the box)

a) A
b) C
c) You could stop training early or go get data.

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Assignment 1: Video Quality Inference

14. [3 points]: In Hands-On 3 and Assignment 1 you computed a segment-download-rate feature from encrypted Netflix traffic. (a) How can you find segment boundaries without decrypting anything? (b) Why do segments carry information about resolution?

(Answer inside the box)

You essentially check to see if there are bursts of small upstream requests and bursts of large downstream responses. b) they carry information because they contain bytes that match up the video, hence related to video resolution. Needs to inform player what resolution to use

15. [4 points]: A student's Assignment 1 classifier scores 75% accuracy with a random train/test split, but about 30% when every session from one household is held out for testing. (a) What does this suggest the model learned? (b) An ISP wants to deploy the model in new homes. Which accuracy is the better estimate, and name one change that would help.

(Answer inside the box)

a) The model probably learned features related to specific household rather the actual network features. b) ~~The~~ The 30% accuracy is a better measure that one change we could do is drop household and other non related information.

Feedback

16. [3 points]: Interest (1=Boring!; 10=Amazing!): 7 Difficulty (1=Too easy; 10=Too hard): 6

One thing you like. One suggestion for improvement:

(Answer inside the box)

Make ~~boxes~~ boxes bigger PLEASE!!! More time for hands on. Delay non-related qus to after class or OH

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